

Research Direction Proposal

Problem and contribution

Statistical tasks used to rely a lot on deterministic methods. Since the advent and broad use of LLM-based tools, data imputation has been done using data agents. However, the trust in these non-deterministic tools depends both on the results and the underlying assumptions made by the data agent. This project compares a data agent across a large number of datasets with values being MAR (missing at random), MCAR (missing completely at random) or MNAR (missing not at random), respectively.

Much of the recent work in the field (David Donoho, 2017; Sun Ming et al., 2025) evaluated systems based on task performance, often overlooking whether these systems can correctly interpret statistical assumptions. This raises an important gap, namely that current evaluations tend to measure output correctness, but not whether AI systems can detect incorrect assumptions provided by users.

This project is motivated by the question: How can we assess trust in human-AI collaboration in statistical tasks via the agent's ability to recognize incorrect assumptions?

Research question (descriptive):

To what extent can LLM-based data agents accurately diagnose the underlying missing data mechanisms (MCAR, MAR, MNAR) and autonomously apply statistically appropriate imputation methods when presented with incomplete datasets?

Methodology

This project uses an empirical evaluation approach. The main goal is to test whether an LLM-based agent can identify missing data mechanisms when given datasets with known missingness patterns. The study consists of three main stages:

1. Dataset selection
 - a. We select a set of classical datasets from the UCI Machine Learning Repository, which contains hundreds of widely used datasets from various fields, like healthcare, finance, and social science.
2. Missing data construction
 - a. For each dataset, we either use datasets with existing missing values (if available), or introduce controlled missingness (MCAR, MAR, MNAR)
 - b. Each mechanism is applied several times with different missing rates and random seeds, creating about 100 datasets per mechanism.
3. Agent evaluation
 - a. The agent analyzes each dataset and identifies the missing data mechanism, providing an explanation.

Data Sources and Limitations

The main data source is the UCI Machine Learning Repository, a popular resource for comparing machine learning methods. We focus on “classic” datasets because they are well-documented, frequently used in research and allow reproducibility.

However, there are limitations as many datasets are relatively clean, true missingness mechanisms are often not documented and some preprocessing is required.

To tackle this, we combine real datasets with synthetically generated missingness, providing both realism and control.

Key Assumptions

This study makes several assumptions:

1. Simulated missingness is a reasonable proxy. We assume that standard methods for generating MCAR, MAR, and MNAR reflect real-world patterns.
2. Model outputs reflect reasoning. We assume that the agent’s explanations give insight into its reasoning, although this may not always be the case.
3. Classic datasets are sufficient for evaluation. While they are widely used, they might not capture all real-world complexities. These assumptions are common but will be acknowledged as limitations.

Experimental Design

We construct datasets as follows:

- 10–20 base datasets with missing rates of 10%–50%, using multiple random seeds and the three mechanisms (MCAR, MAR, MNAR)

This results in ~100 datasets per mechanism with ~300 total experimental cases.

For each dataset, the agent is asked to describe the missing data pattern, identify the mechanism and explain its reasoning. We record both predictions and explanations.

Evaluation Approach

We assess the agent using three types of metrics:

1. Classification accuracy
 - a. This checks if the agent correctly identifies MCAR, MAR, or MNAR.
2. Consistency across prompts
 - a. We test the same dataset under different prompt conditions (e.g., neutral vs. misleading) to see if answers change.
3. Reasoning quality
 - a. We evaluate qualitatively whether the explanation shows a correct statistical understanding. This combined approach helps us tell apart correct reasoning and correct answers by chance or prompt bias

Results to date

We have screened the relevant literature on missing value imputation with data agents for MAR, MCAR and MNAR and collected existing datasets which we will use in our project to measure the correctness of imputed values and assumptions. The current dataset collection consists of 100 datasets from various domains, such as health, business, social sciences and biology. The data sets were acquired from UCI using the filters "202 <= Instances <= 15000" and "0 <= Features <= 50". Results were sorted by views descending, using the first 100 results.

Annex

Planned paper outline

1. Abstract (~0.25 pages)
 - a. Concisely summarize the problem, benchmark approach, key findings, and implications.
2. Introduction (~1 page)
 - a. Establish the context, gap, and contribution.
 - b. Explicitly state research questions and outline the paper's structure.
3. Background and Related Literature (1-2 pages)
 - a. Thematically synthesize substantive and methodological literature, avoiding one-by-one summaries.
4. Research Design and Methods (1-2 pages)
 - a. Detail dataset provenance and analytical methods (RMSE, LLM-as-a-Judge).
 - b. Justify methodology and explicitly disclose any AI tool usage.
5. Results (2-4 pages)
 - a. Objectively report trends, effect sizes, and null results using tables/figures without premature interpretation.
6. Discussion (1.5-3 pages)
 - a. Interpret findings against theoretical framing and discuss implications for human-AI trust.
7. Limitations and Future Research (0.5-1 page)
 - a. Critically evaluate internal, external, and evaluation validity. Propose concrete future research directions.
8. Conclusion (0.5-1 page)
 - a. Synthesize contributions without introducing new claims.

Timeline

Dataset Collection (Week 1-2)

- Collect 100 classic datasets (e.g., UCI Repository)

- Clean and standardize formats
- Create multiple dataset variants

Goal: Build a reliable base dataset pool

Missingness Simulation (Week 3-4)

Inject:

- MCAR (random)
- MAR (depends on observed data)
- MNAR (depends on hidden values)
- Generate ~100 datasets per type

Evaluation (Week 5-6)

Quantitative:

- MSE
- Distribution similarity

Qualitative:

- Code
- Explanations
- Assumption awareness

Output: 3-Level Trust Classification

Writing (Week 7-8)

- Analyze patterns
- Connect to literature
- Discuss limitations

Final Output: Complete research paper

Seed References & Prior Work

Donoho (2017): Highlights the central role of statistical thinking in data science, emphasizing that data analysis is not merely about computation but fundamentally about understanding underlying assumptions. This principle grounds the project's focus on measuring an agent's "assumption-awareness" rather than just output correctness.

Sun et al. (2025, Survey on LLM-based Agents): Provides a comprehensive overview of LLM agents in data science tasks. Crucially, this survey identifies a major gap that this project aims to fill: current evaluations predominantly focus on mechanical task performance, leaving deeper statistical reasoning largely unassessed.

LLMs for Missing Data Imputation (2026): Investigates the specific behavioral quirks of LLMs in missing data tasks, including hallucination effects and logical inconsistencies. This literature

empirically supports the project's core hypothesis: generating a "correct-looking" imputed number does not necessarily imply that the agent possesses correct statistical understanding.

Jäger et al. (2021, A Benchmark for Data Imputation Methods): Establishes a comprehensive benchmark framework for evaluating data imputation algorithms by systematically inducing various missingness mechanisms (MCAR, MAR, MNAR) at different ratios across multiple datasets. This paper provides the direct methodological blueprint and rigorous justification for my automated data generation pipeline (using artificial amputation), ensuring the evaluation testbed is standardized and possesses absolute ground truth.